The chain

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Chapter 03 Stage 3 of 14 High concentration · Moderately hard to substitute

Chemicals, Gases and Photoresist

Japanese companies held 78 percent of world photoresist sales in 2023. Supply of chip chemicals, gases and metals has been disrupted three times since 2019.

Specimen 03 · Chemicals, Gases and Photoresist

A specialty gas cabinet with its door off, and a bulk liquid cylinder standing behind it. The cabinet is about 1 m wide and 1.9 m tall and holds three gas cylinders. The drawing is to scale. The steel-gray panel on the back wall holds the valves, regulator and gauges that control the gas flow.

In plain terms

Photoresist is a coating that changes wherever light hits it. A machine spins a coat of it, thinner than a soap bubble, across the wafer. Light shines through a stencil onto the coat, and a liquid wash then takes away the parts the light reached. What remains is a pattern, and the next machine cuts that pattern into the wafer beneath, a step called etching. Hundreds of other chemicals, gases and materials lay down layers, strip them off and clean the wafer between steps[1]. The cleaning chemicals are purified until metal and other impurities are down to parts per trillion, the Japanese chemical maker Sumitomo Chemical said in 2021[2]. A chipmaker has to test and approve each new resist in its own factory before using it[3]. Even a resist that passes the first time takes seven to 17 months to approve, the Semiconductor Industry Association wrote in 2023[3]. Japanese companies held 78 percent of world photoresist sales in 2023, by a Fuji Keizai survey that Japan's trade ministry reproduced[4].

In short

In July 2019 Japan began requiring an export license for each contract to sell photoresist, hydrogen fluoride or fluorinated polyimide to South Korea[5]. Japan's hydrogen fluoride exports to South Korea then fell 87.9 percent, but

1 Testimony of the Semiconductor Industry Association (SIA) before the Senate... U.S. Senate Committee on Environment and Public Works, 4 March 2026, p. 6. epw.senate.gov

2 Sumitomo Chemical to Expand Production Capacity for High-Purity Chemicals for... Sumitomo Chemical, 24 August 2021. sumitomo-chem.co.jp

3 PFAS-Containing Photo-Acid Generators Used in Semiconductor Manufacturing. Semiconductor Industry Association, 8 June 2023, p. 22. semiconductors.org

4 METI, Semiconductor and digital industry strategy: future direction... Ministry of Economy, Trade and Industry (Japan), 23 December 2025, p. 8. meti.go.jp

5 The impact of export controls on international trade: evidence from the... Research Institute of Economy, Trade and Industry, 8 May 2023. rieti.go.jp

its photoresist exports did not[5]. Production of all three chemicals rose inside South Korea[5].

China banned exports of the metals gallium, germanium and antimony to the United States in December 2024[6]. In Europe, germanium metal averaged an estimated $4,100 a kilogram in 2025, against $1,392 in 2023, on Argus Media prices reported by the US Geological Survey[7]. In November 2025 China suspended the ban until 27 November 2026[8].

A chip fab draws on hundreds of chemicals, gases and other materials[1]. A shortage of any one of them can stop it. The French gas company Air Liquide alone supplies more than 200 different chemicals and roughly 50,000 cylinders a year[9].

Chokepoint card Chemicals, Gases and Photoresist, stage 3 of 14

Concentration

High

Four to six firms could supply this stage today, and the top two take most of the revenue.

Substitutability

Moderate

This rating is based on photoresist, the hardest input to replace at this stage[10].

Price or market size

$45.8bn

market for wafer fabrication materials in 2025[11]. SEMI's count includes silicon wafers and photomasks[12]

Who leads	JP	**JSR**	A vehicle of the state-backed Japan Investment Corporation bought 84.36% of JSR in April 2024[13]
	JP	**Tokyo Ohka Kogyo**	Resist maker with ¥139,668m of sales in the first half of fiscal 2026[14]
	FR	**Air Liquide**	€2,465m of electronics revenue in 2025, 9.1% of group sales[9]
	US	**Entegris**	Bought the polishing-slurry maker CMC Materials in July 2022[15]
	JP	**JX Advanced Metals**	Says it holds about 65% of the world market for semiconductor sputtering targets as of fiscal 2026[16]
Where it is made	JP	**Japan**	Photoresist, ultra-pure hydrofluoric acid, sputtering targets, polishing slurries
	US	**United States**	Entegris polishing slurries, Qnity Electronics resist, Air Products gases
	FR	**France**	Air Liquide electronics gases and precursors
	KR	**South Korea**	Dongjin resist, with output of resist and hydrogen fluoride rising after 2019
	CN	**China**	Gallium, germanium and antimony refining

1 Testimony of the Semiconductor Industry Association (SIA) before the Senate... U.S. Senate Committee on Environment and Public Works, 4 March 2026, p. 6. epw.senate.gov

5 The impact of export controls on international trade: evidence from the... Research Institute of Economy, Trade and Industry, 8 May 2023. rieti.go.jp

6 商务部公告2024年第46号 关于加强相关两用物项对美国出口管制的公告. Ministry of Commerce of the People's Republic of China, 3 December 2024. mofcom.gov.cn

7 Mineral Commodity Summaries 2026: Germanium. U.S. Geological Survey, 5 February 2026. pubs.usgs.gov

8 商务部公告2025年第72号 公布调整实施商务部公告2024年第46号的决定. Ministry of Commerce of the People's Republic of China, 9 November 2025. mofcom.gov.cn

9 Air Liquide, electronics page. Air Liquide, accessed 9 September 2026. airliquide.com

10 PFAS-Containing Photo-Acid Generators Used in Semiconductor Manufacturing. Semiconductor Industry Association, 8 June 2023, p. 21. semiconductors.org

11 Global Semiconductor Materials Market Revenue Reaches Record $73.2 Billion in... PR Newswire, 12 May 2026. prnewswire.com

12 Materials Market Data Subscription (MMDS). SEMI, accessed 20 October 2026. semi.org

13 Announcement Regarding Result of Tender Offer for JSR Corporation (Securities... Japan Investment Corporation, 17 April 2024. jiccapital.co.jp

14 Consolidated Financial Results for the Second Quarter (Interim) of the Fiscal... Tokyo Ohka Kogyo, 6 August 2026, p. 4. tok.co.jp

15 ENTEGRIS INC, Form 8-K current report for the period ended 2022-07-06. U.S. Securities and Exchange Commission, 6 July 2022. sec.gov

16 Quick Guide to JX Advanced Metals. JX Advanced Metals, accessed 9 September 2026. jx-nmm.com

Why substitution is possible

Each formula is tuned to the layers it prints and has to pass qualification in the fab [10]. So a new supplier has to develop its version together with the fab, one layer at a time [3]. When Japan required individual export licenses for resist and two other chemicals sold to South Korea in 2019, South Korea bought more hydrogen fluoride from the United States and Taiwan and more resist from Belgium [5].

Where China stands

In March 2026 the Chinese resist maker Hubei Dinglong said China had yet to reach large-scale production of advanced resist for 248 nm, 193 nm or extreme-ultraviolet printing [17]. Since August 2023 China has required licenses to export gallium and germanium [18].

Where the US stands

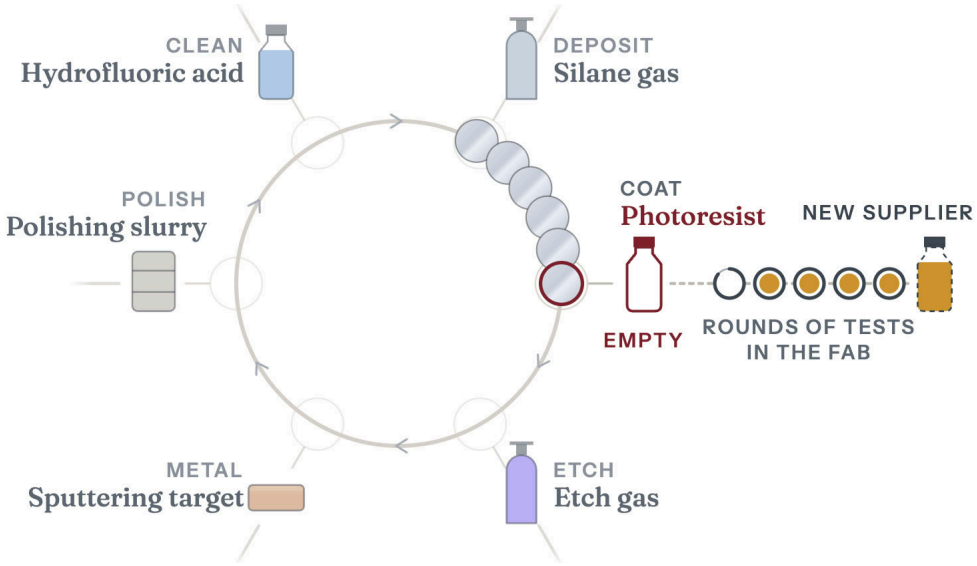
In photoresist, American companies ranked a distant second to Japanese companies by sales in 2023 [4]. In 2021 JSR of Japan agreed to buy Inpria, an Oregon-based US maker of resist for extreme ultraviolet [19].

01

How it works

Figure 03.1 · Explainer, 25 seconds

Why one missing material can stop a fab



1. **One layer.** A chip is built up in layers. For each layer the wafer goes around a loop of steps like this one, and each step uses its own chemical, gas or metal.
2. **Many wafers.** A fab runs many wafers at once, so every material is always in use. Deliveries keep each stock topped up.
3. **One runs out.** If one material stops arriving, the fab works through its stock.
4. **The fab stops.** When the stock runs out, wafers pile up at that step, and the rest of the fab stands idle.

3 PFAS-Containing Photo-Acid Generators Used in Semiconductor Manufacturing. Semiconductor Industry Association, 8 June 2023, p. 22. semiconductors.org

4 METI, Semiconductor and digital industry strategy: future direction... Ministry of Economy, Trade and Industry (Japan), 23 December 2025, p. 8. meti.go.jp

5 The impact of export controls on international trade: evidence from the... Research Institute of Economy, Trade and Industry, 8 May 2023. rieti.go.jp

10 PFAS-Containing Photo-Acid Generators Used in Semiconductor Manufacturing. Semiconductor Industry Association, 8 June 2023, p. 21. semiconductors.org

17 湖北鼎龙控股股份有限公司2025年年度报告摘要. CNINFO, 27 March 2026, p. 2. static.cninfo.com.cn

18 商务部 海关总署公告2023年第23号 关于对镓、锗相关物项实施出口管制的公告. Ministry of Commerce of the People's Republic of China, 3 July 2023. mofcom.gov.cn

19 JSR Agrees to Acquire EUV Pioneer Inpria Corporation. JSR Corporation, 17 September 2021. jsr.co.jp

5. **A new supplier.** A new supplier's resist goes through round after round of tests in the fab. Even one that passes the first time takes seven to 17 months to approve.

Note: Simplified diagram. Six steps and their materials stand for the many a fab uses, and each real layer uses its own mix.

The resist has to change only where the light lands. It also has to work with very little light, so the scanner, the machine that projects the pattern, can expose each patch of the wafer in milliseconds.

Since the 1990s the standard has been chemically amplified resist. In resists for light with wavelengths of 248 nm and 193 nm, a photon absorbed in the film can free an acid molecule, and little more happens until the wafer is baked. In the bake, each acid molecule sets off hundreds of reactions around itself that turn the film soluble, so a little light does a lot of chemistry[20]. The wash then takes away the exposed film and leaves the rest.

The 248 nm light comes from a krypton fluoride (KrF) laser. The 193 nm light comes from an argon fluoride (ArF) laser. Immersion scanners also pass it through a layer of water, which sharpens the image.

Chemically amplified resist works less well with extreme ultraviolet (EUV) light (see Lithography). An absorbed EUV photon instead knocks electrons loose, and a cascade of lower-energy secondary electrons makes the chemical changes in the resist[21]. A 13.5 nm EUV photon carries about fourteen times the energy of a 193 nm one, so the same dose of light arrives as far fewer photons. With fewer photons, random variation in where each one lands matters more. It shows up as ragged line edges and missing contact holes, the tiny holes that link one layer of a chip to the next.

Resist makers responded with metal-oxide resists, which are built on clusters with a tin-oxide core and absorb more of the EUV light than chemically amplified resists[22]. In 2021 the Japanese resist maker JSR agreed to buy Inpria, an Oregon-based US maker of these resists[19].

The other chemicals and gases put material on the wafer, take it off or keep it clean. Bulk gases such as nitrogen come by pipeline or from plants built at the fab, and specialty gases come in cylinders[9]. Silane gas carries the silicon for new layers into the chamber that holds the wafer. A plasma splits nitrogen trifluoride, and the fluorine it frees scrubs the chamber walls clean between wafers. Hydrofluoric acid cleans the wafer itself. Polishing slurries grind each new layer flat before the next goes on.

Sputtering targets are slabs of pure metal. A plasma knocks atoms off them, and the atoms settle on the wafer as thin metal films, including the barrier and seed layers for copper wiring[23]. Electroplating then fills in the copper wiring itself[24]. Filters and wafer carriers keep contamination out.

9 Air Liquide, electronics page. Air Liquide, accessed 9 September 2026. airliquide.com

19 JSR Agrees to Acquire EUV Pioneer Inpria Corporation. JSR Corporation, 17 September 2021. jsr.co.jp

20 PFAS-Containing Photo-Acid Generators Used in Semiconductor Manufacturing. Semiconductor Industry Association, 8 June 2023, p. 6. semiconductors.org

21 How Scientists Are Accelerating the Development of Next-Gen Microelectronics. Lawrence Berkeley National Laboratory, 21 August 2023. newscenter.lbl.gov

22 Inpria, company website. Inpria, accessed 9 September 2026. inpria.com

23 Applied Materials, PVD technology page. Applied Materials, accessed 9 September 2026. appliedmaterials.com

24 SABRE Product Family. Lam Research, 11 August 2017. lamresearch.com

02

Variants and trade-offs

Resist class	Wavelength	Where it is used	Limit
i-line novolak	365 nm	Packaging, mature nodes	Resolution
KrF chemically amplified	248 nm	Implant layers, thick films	Resolution
ArF immersion chemically amplified	193 nm	Most critical layers below 28 nm	Needs several exposures per layer
EUV chemically amplified	13.5 nm	Leading-edge single-exposure layers	Shot noise from too few photons
EUV metal-oxide	13.5 nm	High-NA tools, tightest pitches	Cost, defects, outgassing

Each resist class is made for one kind of lithography, such as EUV, ArF immersion, ArF, KrF or i-line[25].

03

Who makes it

Japanese companies sell most of the world's photoresist, by two different counts. In a January 2021 brief, Georgetown University's Center for Security and Emerging Technology (CSET) put Japan's share of the semiconductor photoresist market at 90 percent, citing a 2019 estimate[25]. The brief put most of the rest with US and South Korean firms[25]. In a December 2025 strategy presentation, Japan's Ministry of Economy, Trade and Industry (METI) published figures from a separate survey by Fuji Keizai, a Japanese market research firm[4]. That survey counts sales by company nationality and puts Japanese companies at 77 percent in 2021 and 78 percent in 2023, with American and South Korean companies next[4].

Figure 03.2

Japanese companies held 78 percent of photoresist sales in 2023

Photoresist sales share by company nationality, 2023



Source: METI, December 2025, from Fuji Keizai data [4]

CSET's January 2021 brief lists five Japanese resist makers: JSR, Tokyo Ohka Kogyo, Shin-Etsu Chemical, Fujifilm and Sumitomo Chemical[26]. It also lists Dongjin of South Korea, four Chinese firms and three firms it counts as American: DuPont, EMD Performance Materials and Inpria[26]. Qnity Electronics, the former DuPont electronics business, says in its 2025 annual report that it offers lithography materials for every wavelength, including EUV[27].

4 METI, Semiconductor and digital industry strategy: future direction... Ministry of Economy, Trade and Industry (Japan), 23 December 2025, p. 8. meti.go.jp

25 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 59. cset.georgetown.edu

26 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 56. cset.georgetown.edu

27 Qnity Electronics, Inc., Form 10-K annual report for the period ended... U.S. Securities and Exchange Commission, 26 February 2026. sec.gov

In its 2025 annual report summary, the Chinese firm Hubei Dinglong said China had yet to reach large-scale production of advanced resist for 248 nm, 193 nm or EUV printing[17]. But it said three of its own resists for 248 nm and 193 nm printing had moved into stable volume supply[28].

A Japanese state-backed fund, Japan Investment Corporation, took control of JSR in April 2024. A company the fund controls bought 84.36 percent of JSR's shares at ¥4,350 each in an offer to all shareholders that closed on 16 April 2024[13]. JSR's shares were delisted from the Tokyo Stock Exchange on 25 June 2024[29].

Two Japanese suppliers tie their growth to demand for AI chips. Tokyo Ohka Kogyo sold ¥139,668 million in the first half of fiscal 2026, up 25.1 percent, as strong demand for AI servers in data centers raised demand for chips[14]. Fujifilm planned to raise its polishing-slurry capacity at Kumamoto in Japan by about 30 percent from January 2025, to meet demand driven by AI chips[30]. It also scheduled new production and quality-evaluation equipment for EUV resist at Shizuoka in Japan and Pyeongtaek in South Korea to start in October 2025[31].

World sales of wafer fabrication materials reached $45.8 billion in 2025, and sales of packaging materials reached $27.4 billion[11]. SEMI counts silicon wafers and photomasks among wafer fabrication materials, along with resists, gases, chemicals and polishing materials[12].

In electronic gases, CSET named Merck, Air Products and Air Liquide as the market leaders in January 2021[32]. Air Liquide's electronics business sold €2,465 million in 2025, 9.1 percent of group revenue[9]. SK Specialty has made nitrogen trifluoride since 2001, when it became the first company to produce the gas in South Korea[33]. The Japanese firm Kanto Denka Kogyo refines its own fluorine gas and turns it into specialty gases for deposition, etching and cleaning[34]. Stella Chemifa, also Japanese, makes ultra-high-purity hydrofluoric acid for chipmaking[35]. Another Japanese firm, Morita Chemical Industries, sells an electronic-grade hydrofluoric acid[36].

A few firms held most of the markets for wet chemicals and polishing slurries in 2019. CSET's January 2021 brief, drawing on a 2019 report by the market research firm Mordor Intelligence, put five firms, KMG Chemicals, BASF, Avantor, Honeywell and Kanto Chemical, at more than 60 percent of the wet chemicals market[32]. Citing a 2019

9 Air Liquide, electronics page. Air Liquide, accessed 9 September 2026. airliquide.com

11 Global Semiconductor Materials Market Revenue Reaches Record $73.2 Billion in... PR Newswire, 12 May 2026. prnewswire.com

12 Materials Market Data Subscription (MMDS). SEMI, accessed 20 October 2026. semi.org

13 Announcement Regarding Result of Tender Offer for JSR Corporation (Securities... Japan Investment Corporation, 17 April 2024. jiccapital.co.jp

14 Consolidated Financial Results for the Second Quarter (Interim) of the Fiscal... Tokyo Ohka Kogyo, 6 August 2026, p. 4. tok.co.jp

17 湖北鼎龙控股股份有限公司2025年年度报告摘要. CNINFO, 27 March 2026, p. 2. static.cninfo.com.cn

28 湖北鼎龙控股股份有限公司2025年年度报告摘要. CNINFO, 27 March 2026, p. 8. static.cninfo.com.cn

29 Announcement of Delisting of Shares of JSR Corporation. JSR Corporation, 24 June 2024. jsr.co.jp

30 Fujifilm to Enhance its Production Capacity of Advanced Semiconductor Material... Fujifilm, 5 December 2024. fujifilm.com

31 Fujifilm Launches EUV Resist and EUV Developer. Fujifilm, 29 October 2024. fujifilm.com

32 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 60. cset.georgetown.edu

33 SK specialty, company website. SK specialty, accessed 18 October 2026. skspecialty.com

34 Specialty Gases. Kanto Denka Kogyo, accessed 18 October 2026. kantodenka.co.jp

35 High-purity Chemical Business. Stella Chemifa, accessed 18 October 2026. stella-chemifa.co.jp

36 Electronic grade Hydrofluoric acid. Morita Chemical Industries, accessed 18 October 2026. morita-kagaku.co.jp

report by the market research firm Market Reports World, it put DuPont and Cabot Microelectronics at about 56 percent of a $790 million market for polishing slurries[25]. Counting by company nationality, the same Fuji Keizai survey, as published by METI, puts Japanese companies at 51 percent of polishing-slurry sales and 59 percent of high-purity cleaning-liquid sales in 2023[4]. Both shares were 49 percent in 2021[4].

The slurry businesses of both leaders CSET named have since moved to other companies. Cabot Microelectronics renamed itself CMC Materials in October 2020[37]. The American firm Entegris completed its purchase of CMC Materials on 6 July 2022[15]. DuPont separated its electronics business into an independent company, Qnity Electronics, on 1 November 2025[38]. Qnity sells polishing slurries and pads[39].

In sputtering targets for semiconductors, JX Advanced Metals puts its own share of the world market at about 65 percent as of fiscal 2026[16].

04

The chokepoint

Supply at this stage has been disrupted three times since 2019, twice by governments and once by a war.

In July 2019 Japan put photoresist, hydrogen fluoride and fluorinated polyimide under individual export licenses for South Korea, so each export contract needed its own license[5]. South Korea's chip industry depended heavily on Japan for all three chemicals, and Japanese firms supplied more than 90 percent of South Korea's imports of two of them[5]. Japan's hydrogen fluoride exports to South Korea then fell 87.9 percent[5]. Japanese photoresist exports to South Korea did not fall, possibly in part because Japan allowed three-year bulk licenses for some resist sales in December 2019[5].

South Korea bought more hydrogen fluoride from the United States and Taiwan and more photoresist from Belgium[5]. Production of all three chemicals rose inside South Korea[5]. The rise came both from South Korean firms and from Japanese companies' affiliates in South Korea[5].

Researchers at Japan's Research Institute of Economy, Trade and Industry (RIETI) found that Japanese hydrogen fluoride exports to the United States rose[5]. They say some Japanese supply may have reached South Korea by way of the United States[5].

4 METI, Semiconductor and digital industry strategy: future direction... Ministry of Economy, Trade and Industry (Japan), 23 December 2025, p. 8. meti.go.jp

5 The impact of export controls on international trade: evidence from the... Research Institute of Economy, Trade and Industry, 8 May 2023. rieti.go.jp

15 ENTEGRIS INC, Form 8-K current report for the period ended 2022-07-06. U.S. Securities and Exchange Commission, 6 July 2022. sec.gov

16 Quick Guide to JX Advanced Metals. JX Advanced Metals, accessed 9 September 2026. jx-nmm.com

25 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 59. cset.georgetown.edu

37 CMC Materials, Inc., Form 8-K current report for the period ended 2020-09-28... U.S. Securities and Exchange Commission, 1 October 2020. sec.gov

38 DuPont Completes Separation of Qnity Electronics. DuPont, 3 November 2025. investors.dupont.com

39 Chemical Mechanical Planarization (CMP) Materials for Semiconductor. Qnity Electronics, accessed 9 September 2026. qnityelectronics.com

Japan ended the individual licenses on 23 March 2023, when it put exports of the three chemicals to South Korea under its bulk export license system[40].

Russia's invasion of Ukraine in 2022 put chipmakers' supply of neon at risk. Before the invasion, Ukraine supplied about 70 percent of the world's neon and 90 percent of the semiconductor-grade neon used by US industry[41]. The fighting forced at least two of Ukraine's three major gas producers, Cryoin and Ingas, to shut their operations in Odesa and Mariupol[41]. That halted a significant part of the world's exports of neon, krypton and xenon[41]. Gas mixtures containing neon fill the lasers used in lithography, but EUV scanners use a tin light source instead[42]. Many chipmakers had added to their gas stockpiles after Russia annexed Crimea in 2014[41]. Samsung, SK hynix, Renesas and Rohm reported adequate neon supplies from China[41].

For chipmaking materials, China's export controls cover metals. On 3 July 2023 China announced that exports of gallium and germanium would need licenses from 1 August[18]. On 3 December 2024 it said it would in principle grant no licenses to export gallium, germanium, antimony or superhard materials to the United States[6]. On 9 November 2025 it suspended that ban until 27 November 2026[8]. In February 2026 the US Geological Survey put China's share of world production of primary low-purity gallium at 99 percent[43].

The White House set out a trade deal with China on 1 November 2025[44]. It said that under the deal China would issue general licenses for exports of gallium, germanium and antimony for the benefit of US end users and their suppliers[44].

Most gallium used in the United States goes into chips, and both metals are recovered as byproducts of processing other minerals. Of the gallium used in the United States in 2025, 73 percent went into integrated circuits built on gallium arsenide or gallium nitride[43]. Another 26 percent went into optoelectronic devices such as LEDs and laser diodes[43]. In the United States, germanium goes mainly into fiber optics, infrared optics, semiconductors and solar cells, and radiation detectors[7]. Gallium is recovered mostly as a byproduct of processing bauxite, the ore that aluminum is made from[43]. Germanium is recovered from certain zinc smelter residues and from coal ash[45]. Because both metals are economical to produce only as byproducts, their supply may not rise much when their prices do[45].

In Europe, germanium metal averaged an estimated $4,100 a kilogram in 2025, against $1,392 in 2023, on Argus Media prices reported by the US Geological Survey[7]. US imports of the metal fell an estimated 67 percent in 2025 because of China's ban[7]. In

6 商务部公告2024年第46号 关于加强相关两用物项对美国出口管制的公告. Ministry of Commerce of the People's Republic of China, 3 December 2024. mofcom.gov.cn

7 Mineral Commodity Summaries 2026: Germanium. U.S. Geological Survey, 5 February 2026. pubs.usgs.gov

8 商务部公告2025年第72号 公布调整实施商务部公告2024年第46号的决定. Ministry of Commerce of the People's Republic of China, 9 November 2025. mofcom.gov.cn

18 商务部 海关总署公告2023年第23号 关于对镓、锗相关物项实施出口管制的公告. Ministry of Commerce of the People's Republic of China, 3 July 2023. mofcom.gov.cn

40 Circular Notice on Partial Amendment to the Circular Notice on the Operation of... Ministry of Economy, Trade and Industry (Japan), 23 March 2023. web.archive.org

41 Russia's Invasion of Ukraine Impacts Gas Markets Critical to Chip Production. Center for Strategic and International Studies, 14 March 2022. csis.org

42 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 97. cset.georgetown.edu

43 Mineral Commodity Summaries 2026: Gallium. U.S. Geological Survey, 5 February 2026. pubs.usgs.gov

44 Fact Sheet: President Donald J. Trump Strikes Deal on Economic and Trade... The White House, 1 November 2025. whitehouse.gov

45 Quantifying Potential Effects of China's Gallium and Germanium Export... U.S. Geological Survey, 15 October 2024, p. 11. pubs.usgs.gov

October 2024 the US Geological Survey estimated that a total Chinese ban on gallium and germanium exports could cut US GDP by $3.4 billion[46].

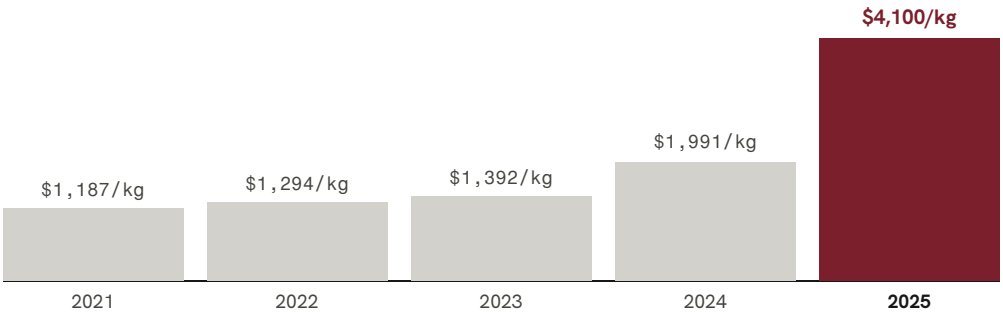
A slower risk is the European Union's proposed restriction on PFAS, the long-lived fluorinated chemicals. These chemicals are used throughout a fab, from some etch gases to seals, filters and the plastic linings of chemical tanks and pipes. Most of the photoacid generators that free the acid in resist contain PFAS too[47].

Each resist made without PFAS would need full qualification in the chipmaker's fab[3]. The Semiconductor Industry Association wrote in 2023 that photoresist needs the most rigorous qualification of any chemical a chipmaker uses, because each resist is tuned to the layers it prints[10]. The resist maker adjusts its formula over rounds of tests in the fab[3]. A fab could not qualify all the new resists at once, mainly because testing several together can muddle the yield results[3]. So some would have to be qualified one after another[3].

Figure 03.3

Germanium in Europe rose to an estimated $4,100 a kilogram in 2025

Germanium metal, annual average price in Europe. China required export licenses from August 2023, banned shipments to the US in December 2024 and suspended the ban from November 2025 to 27 November 2026



Source: USGS Mineral Commodity Summaries, February 2026 (Argus Media European prices, 2025 estimated) [7]

The EU proposal may still let chipmakers keep using PFAS where the risks can be controlled. The national authorities of Denmark, Germany, the Netherlands, Norway and Sweden prepared the proposal and updated it after evaluating more than 5,600 comments[48]. ECHA, the EU chemicals agency, published the updated proposal on 20 August 2025[48]. The updated proposal assessed options that would let electronics and semiconductor makers keep using PFAS where the risks can be controlled[48].

In March 2026 ECHA's two scientific committees backed an EU-wide restriction with targeted derogations, or exemptions[49]. One committee gave its backing in a final

3 PFAS-Containing Photo-Acid Generators Used in Semiconductor Manufacturing. Semiconductor Industry Association, 8 June 2023, p. 22. semiconductors.org

7 Mineral Commodity Summaries 2026: Germanium. U.S. Geological Survey, 5 February 2026. pubs.usgs.gov

10 PFAS-Containing Photo-Acid Generators Used in Semiconductor Manufacturing. Semiconductor Industry Association, 8 June 2023, p. 21. semiconductors.org

46 Quantifying Potential Effects of China's Gallium and Germanium Export... U.S. Geological Survey, 15 October 2024, p. 9. pubs.usgs.gov

47 PFAS-Containing Photo-Acid Generators Used in Semiconductor Manufacturing. Semiconductor Industry Association, 8 June 2023, p. 4. semiconductors.org

48 ECHA publishes updated PFAS restriction proposal. European Chemicals Agency, 20 August 2025. echa.europa.eu

49 ECHA supports PFAS restriction with targeted derogations. European Chemicals Agency, 26 March 2026. echa.europa.eu

opinion and the other in a draft[49]. Their evaluation is due to finish by the end of 2026[49]. The European Commission will decide in consultation with the member states[48].

Four questions to answer from memory

1. What does photoresist do? see How it works
2. Which country's companies sell most of the world's photoresist? see Who makes it
3. Why is it slow to switch to a new photoresist supplier? see The chokepoint
4. Which two metals has China required a license to export since August 2023? see The chokepoint

Four things to remember

- Photoresist holds the pattern printed by light so the next step can etch it into the wafer.
- Japanese companies held 77 percent of world photoresist sales in 2021 and 78 percent in 2023.
- A chipmaker has to approve each new resist in its own fab, and even one that passes the first time takes seven to 17 months.
- China has required licenses to export gallium and germanium since August 2023.

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A company filings, governments, standards bodies, universities and research institutions; B SemiAnalysis, Reuters, the Financial Times and Nikkei Asia.

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